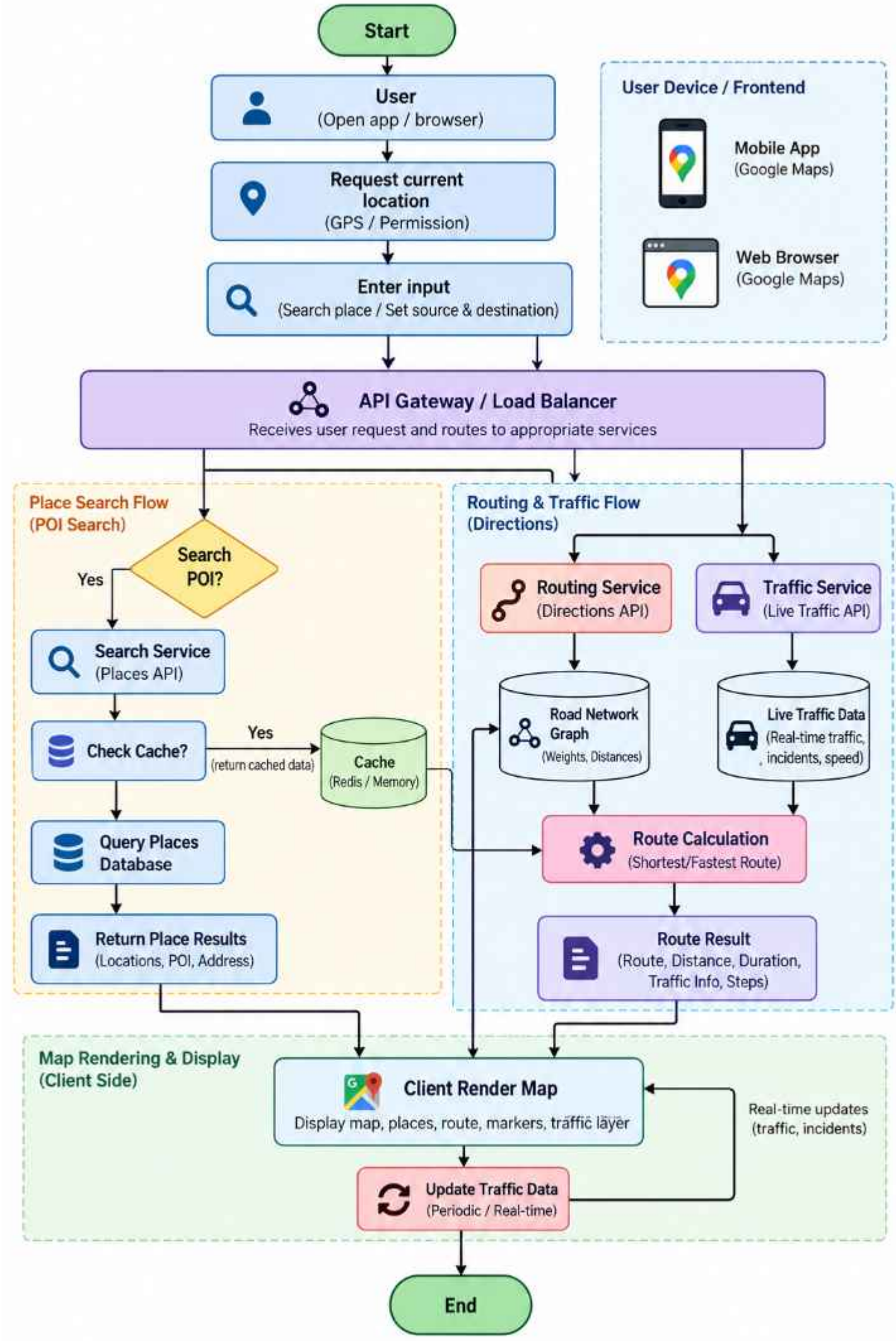

Start
User
(Open app / browser)
Request current location
(GPS / Permission)
Enter input
(Search place / Set source & destination)
User Device / Frontend
Mobile App
(Google Maps)
Web Browser
(Google Maps)
API Gateway / Load Balancer
Receives user request and routes to appropriate services
Place Search Flow
(POI Search)
Search POI?
Yes
Search Service
(Places API)
Check Cache?
Yes
(return cached data)
Cache
(Redis / Memory)
Query Places Database
Return Place Results
(Locations, POI, Address)
Routing & Traffic Flow
(Directions)
Routing Service
(Directions API)
Traffic Service
(Live Traffic API)
Road Network Graph
(Weights, Distances)
Live Traffic Data
(Real-time traffic, incidents, speed)
Route Calculation
(Shortest/Fastest Route)
Route Result
(Route, Distance, Duration, Traffic Info, Steps)
Map Rendering & Display
(Client Side)
Client Render Map
Display map, places, route, markers, traffic layer
Real-time updates
(traffic, incidents)
Update Traffic Data
(Periodic / Real-time)
End